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The paleomagnetism of the Upper Cretaceous magmatic rocks in the Banat area of South Carpathians: tectonic implications

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ABSTRACT

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Forty-four sites were sampled in Upper Cretaceous magmatic rocks in the western part of the Southern Carpathians (Banat area) for a paleomagnetic study. After detailed AF demagnetization, only 10 sites yielded primary remanent magnetizations, the directions of which are similar to those identified in Senonian magmatic rocks from the Apuseni Mountains and the Haţeg and Rusca Montană Basins. The mean paleomagnetic pole for these three areas is compared with those from coeval formations of stable Europe and stable Africa. This comparison shows a large clockwise rotation, both with respect to Africa and Europe, and a northward transport to reach their present-day position during the Tertiary. The relations with adjacent zones of the Tethys mobile belt are also discussed.

Introduction

The Apuseni–Srednogorie magmatic belt of the Balkan Peninsula is a prominent structural zone of the Alpine tectonic system. It is characterized by volcanic and intrusive rocks which were emplaced mostly during the Late Cretaceous (Strutinski et al., 1986; Popov, 1987). This arc-shaped magmatic belt (Fig. 1) is about 1000 km long and 30–120 km wide. It stretches from the Apuseni Mountains in Romania, through the western part of the South Carpathians (Banat, Romania) and the Timok zone (eastern Serbia), and then it turns to the east in the Bulgarian Srednogorie Mountains and submerges beneath the Black Sea (Fig. 1).

In Romania, these magmatic rocks outcrop in three major tectonic units: the Internal Dacides and Western Transylvanides in the Apuseni Mountains, and the Middle Dacides in the western part of the Southern Carpathians (Săndulescu, 1984).

Previous paleomagnetic studies in Upper Cretaceous magmatic rocks from the Romanian sector of this belt were performed on samples from Apuseni Mountains (Pătraşcu et al., 1990) and from the Rusca Montană and Haţeg Basins in the western part of the Southern Carpathians (Pătraşcu and Panaiotu, 1990). The results from the two areas are similar, and indicate for the studied areas a large clockwise rotation and an important northward displacement during the Tertiary to reach their present-day positions. A similar rotation was discovered on some remagnetized Mesozoic rocks from the Pădurea Craiului Mountains, situated in the northwestern part of the Apuseni Mountains (Surmont et al., 1990).

This large clockwise rotation is in accordance with some tectonic models of the central Mediterranean, based either on paleomagnetic

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data (Marton, 1987; Marton and Mauritsch, 1990) or geological data (Boccaletti et al., 1974; Burchfiel, 1980; Balla, 1985).

In this paper, we report paleomagnetic results from the Banat sector of this magmatic belt.

Geological setting

In the Banat area (Fig. 2) the Upper Cretaceous magmatic activity developed in two main stages.

In the first stage, large polyphasic plutons were emplaced (Surduc 1, Bocşa 1 and 2) in which there are gabbros, monzodiorites, followed by monzogranitic differentiates. The second stage started with andesitic volcanic rocks, followed by plutons and hypabyssal bodies of granodiorites at Bocşa 3, Ocna de Fier-Dognecea, Ciclova, Sasca Montană (Andrei et al., 1989).

Generally, the granodiorite and granite plutons generated hornfelses and skarns at the contact with adjacent formations (limestones or other

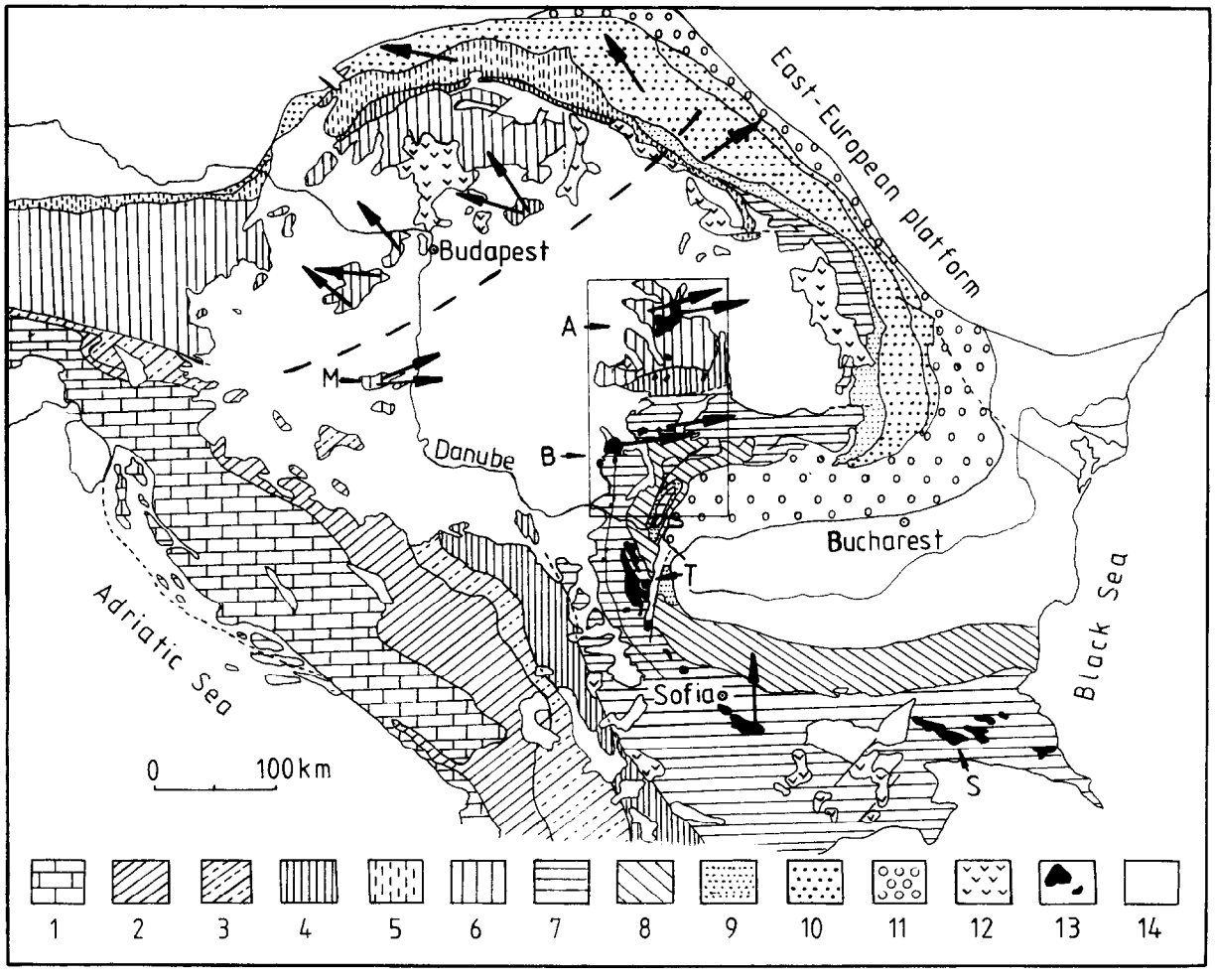


Fig. 1. The location of the region studied (Banat area) in the simplified structural map of central and SE Alpine Europe (after Săndulescu, 1984). Symbols: *1* = Outer Dinarides; *2* = Inner Dinarides; *3* = Palagonides and their equivalents; *4* = Tethyan suture; *5* = Magura units; *6* = Internal Dacides; *7* = Middle Dacides and their equivalents in the Balkan Peninsula; *8* = Marginal Dacides; *9* = Outer Dacides; *10* = Moldavides; *11* = folded molasse; *12* = Neogene volcanic rocks; *13* = Upper Cretaceous magmatic rocks from the Apuseni–Srednogorie magmatic belt; *14* = post-tectonic basins. Area-symbols: *A* = Apuseni Mountains; *B* = Banat; *M* = Mecsek Mountains; *S* = Srednogorie; *T* = Timok. Dashed line = limit between areas with different rotations (taken from Márton, 1987). Post-Jurassic paleomagnetic declinations (arrows) after Márton (1987), Márton and Mauritsch (1990), Surmont et al. (1990) and this study. The boxed area is detailed in Fig. 2.

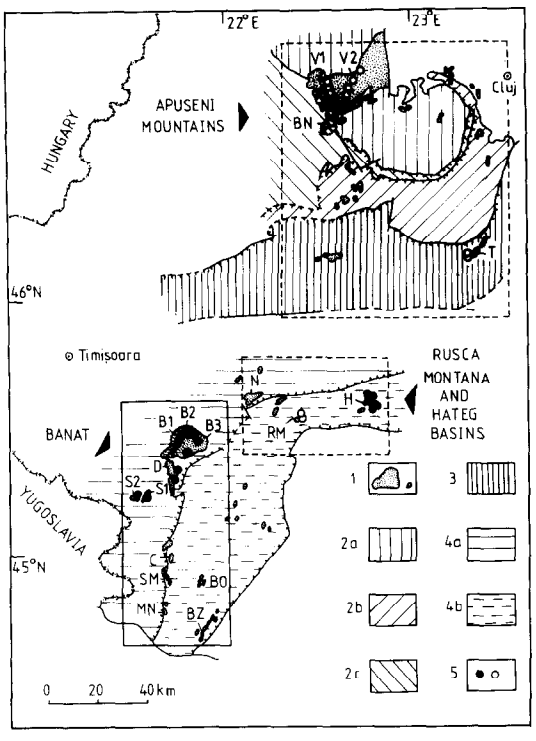


Fig. 2. Sketch of distribution of Upper Cretaceous magmatic rocks in the Apuseni Mountains and the Southern Carpathians (after Russo-Săndulescu et al., 1984). The location of sampling sites used for tectonic interpretations is after Pătraşcu et al. (1990), Pătraşcu and Panaiotu (1990) (dashed boxed areas) and this study (solid boxed area). Symbols: *1* = Upper Cretaceous magmatic rocks; *2* = Internal Dacides (*a* = Bihor Units, *b* = Biharia Units, *c* = Codru Units); *3* = Transylvanides; *4* = Middle Dacides (*a* = Supragetic Units, *b* = Getic Units); *5* = Sampling sites (full circle = normal polarity, open circle = reversed polarity). Sampling area symbols for the Southern Carpathians: *N* = Nădrag; *B1*, *B2*, *B3* = Bocşa; *D* = Dognecea; *S1*, *S2* = Surduc; *C* = Ciclova; *SM* = Sasca Montană; *MN* = Moldova Nouă; *BZ* = Berzasca; *BO* = Bozovici; *RM* = Rusca Montană Basin; *H* = Haţeg Basin. Sampling area symbols for the Apuseni Mountains: *V1*, *V2* = Vlădeasa Mountains; *BN* = Northern Bihor Mountains; *T* = Trascău Mountains.

sedimentary or metamorphic rocks). These plutons penetrate both Getic and Supragetic units from the Middle Dacides (Cioflică, 1989).

K–Ar ages measured during the past decade suggest the following timing of emplacement of these magmatites: Campanian–Maastrichtian in the Poiana Ruscă Mountains (Strutinski et al., 1986); Coniacian–Maastrichtian for Surduc 1, Bocşa 1 and 2 and Maastrichtian–Eocene for Surduc 2 and Bocşa 3 (Russo-Săndulescu et al., 1986); Campanian–Maastrichtian in the southern part of Banat (Soroiu et al., 1986).

The possible temporal extension of magmatism during the Maastrichtian–Eocene in the Bocşa 3 and Surduc 2 areas is based on K–Ar ages on biotite from Russo-Săndulescu et al. (1986). There are two samples with 65 ± 2 Ma and 56 ± 2 Ma in Bocşa 3 and two samples with 62 ± 2 Ma and 55 ± 2 Ma in Surduc 2. Taking into account the fact that biotite shows different degrees of alteration in this areas (Russo-Săndulescu et al., 1986; this study), it is quite probably that the K–Ar ages on biotite are perturbed by the migration of Ar, which was prompted by the alteration (Mitchell et al., 1988). Moreover, in other areas of this magmatic belt, the duration of the magmatism is estimated to be 5–6 Ma, either based on K–Ar ages from the southern part of Banat (Soroiu et al., 1986) or based on K–Ar ages and paleomagnetic data from the Apuseni Mountains (Pătraşcu et al., 1990).

Taking these remarks into account, we think that the extension of the magmatism in our sampling areas in the Paleogene is not well proven, and we consider that the age of all the magmatic rocks used for this paleomagnetic study is probably Campanian–Maastrichtian.

Oriented samples were collected from the following areas (Fig. 2): the Nădrag area in the western part of the Poiana Ruscă Mountains (3 sites), Bocşa (17 sites), Ocna de Fier-Dognecea (6 sites), Surduc (4 sites), Ciclova (4 sites), Sasca Montană (3 sites), Moldova Nouă (3 sites), Berzasca (3 sites) and Bozovici (1 site).

The intrusive bodies have frequently been mineralogically altered. The most important changes can be summarized as follows.

(1) In the gabbros and monzodiorites: olivine is replaced by talc, magnetite and chlorite; pyroxene is altered to green calcic amphibole; and titanomagnetite is partially converted to ilmenite and Ti-poor titanomagnetite.

(2) In the granites and granodiorites: amphibole and biotite are partially altered to chlorite, epidote and sphene; alkali feldspar and plagioclase become turbid and with progressive alteration; and magnetite has been partially converted to hematite.

TABLE 1

Paleomagnetic results from Upper Cretaceous magmatic rocks of the Romanian Carpathians

Site	N/n	D (degrees)	I (degrees)	α_{95} (degrees)	k	ϕ (degrees)	λ (degrees)	δ_p (degrees)	δ_m (degrees)
SOUTHERN CARPATHIANS									
Banat Area									
Bocşa									
B2-11	2/ 8	107.1	22.9	11.4	24	− 3.1	91.2	6.4	12.1
B2-18	3/12	103.0	18.9	2.9	224	− 2.0	95.7	1.6	3.0
B3-2	4/16	100.6	37.2	5.7	42	7.5	89.9	3.9	6.7
B3-27	2/ 8	76.4	40.3	9.6	34	25.5	104.2	7.0	11.0
B3-30	2/ 8	102.0	42.1	1.9	843	9.3	86.4	1.4	2.3
B3-31	3/12	83.9	39.1	5.2	70	19.3	100.0	3.7	5.2
Dognecea									
D-5	3/12	63.2	49.4	2.3	352	39.2	106.0	2.0	3.0
D-13	4/16	83.9	49.9	6.8	73	25.3	92.3	8.8	13.1
Surduc									
S1-22	3/12	49.7	49.9	3.9	124	49.4	114.7	3.5	5.2
S2-21	3/12	48.2	42.3	6.8	42	46.4	122.8	5.2	3.4
Mean	10	83.9	40.9	12.1	17	20.5	98.6	8.9	11.1
	10			13.1	14	21.8	98.8		
Rusca Montană Basin									
RM-115	3/12	247.0	− 27.2	9.5	22	− 26.3	298.6	5.6	10.8
RM-116	4/16	271.4	− 15.1	2.4	549	− 4.5	290.0	1.2	2.4
Haţeg Basin									
H-72	3/12	76.1	53.1	6.1	57	31.8	94.7	5.9	8.5
H-78	3/12	97.3	21.8	2.4	320	3.0	99.7	1.4	2.6
H-82	3/12	46.9	45.7	3.6	113	48.7	122.9	3.0	4.6
H-83	3/12	51.8	34.6	2.9	223	39.9	127.1	1.9	3.3
H-111	3/12	91.3	24.6	4.6	89	8.3	102.8	2.7	5.0
Mean	7	76.2	32.9	16.6	14	22.2	109.3	10.6	15.7
	7			16.2	15	23.4	109.8		
APUSENI MOUNTAINS									
Vlădeasa Mountains									
V1-35	4/16	259.0	− 48.0	3.0	10	− 28.0	279.0	3.1	4.8
V1-36	4/16	257.9	− 41.2	2.5	225	− 25.0	284.0	1.8	3.0
V1-38	4/16	250.0	− 20.4	4.5	69	− 21.3	299.9	2.5	4.7
V1-65	3/12	257.3	− 32.0	3.4	162	− 21.2	289.4	2.2	3.8
V1-66	3/12	268.3	− 11.9	3.4	163	− 5.5	289.7	1.8	3.5
V1-67	3/12	263.4	− 26.2	5.7	59	− 14.5	287.7	3.3	6.1
V1-68	3/12	264.3	− 29.6	4.0	121	− 15.3	285.7	2.4	4.4
V1-69	2/ 8	229.8	− 29.6	2.7	423	− 38.6	312.4	1.6	3.0
V1-70	3/12	278.5	− 38.3	4.6	89	− 10.0	271.6	3.2	5.5
V1-71	3/12	264.2	− 20.1	3.5	153	− 11.5	289.6	1.9	3.7
V2-41	4/16	285.1	− 58.6	2.9	166	− 19.0	254.9	3.2	4.3
V2-76	3/12	259.2	− 39.6	3.4	166	− 23.2	284.4	2.4	4.0
V2-77	3/12	273.0	− 40.6	5.7	58	− 14.7	274.3	4.2	7.0
V2-78	2/ 8	261.3	− 32.9	7.7	52	− 18.8	286.3	5.0	8.8

TABLE 1 (continued)

Site	N/n	D (degrees)	I (degrees)	α_{95} (degrees)	k	ϕ (degrees)	λ (degrees)	δ_p (degrees)	δ_m (degrees)
Northern Bihor Mountains									
BN-32	4/16	245.0	−41.1	4.4	70	−33.8	293.0	3.3	5.4
BN-34	4/16	272.1	−48.5	4.1	84	−19.6	270.1	3.5	5.3
BN-44	2/ 8	247.2	−34.7	6.1	226	−29.3	295.4	4.0	7.0
BN-45	3/12	263.6	−51.9	2.4	331	−27.0	272.9	2.2	3.3
BN-46	4/16	250.3	−46.1	7.5	25	−32.7	285.8	6.1	9.6
Trascău Mountains									
T-48	3/12	253.0	−68.1	6.1	51	−43.5	259.1	8.6	10.3
T-52	3/12	275.3	−31.2	6.6	45	−8.5	277.9	4.1	7.4
Mean	21	260.0	−38.0	6.5	24	−22.0	284.0	5.0	8.0
	21			6.1	28	−22.0	283.0		
Mean of regions									
	38	79.7	37.7	5.5	19	22.0	104.2	3.8	5.1
	38			5.3	20	22.4	103.1		
	3	79.9	37.1	7.7	254	21.6	104.5	5.3	7.2
	3			7.9	245	22.5	103.8		

Site-mean: N/n = samples/specimens.

Area-mean: N/n = number of sites; first row computed from site-mean directions; second row computed from VGPs.

Mean of regions: rows 1 and 2 computed from site-mean directions and VGPs; rows 3 and 4 computed from area-mean directions and VGPs.

Other symbols as in Fisher's statistics (1953). Paleomagnetic data from the Apuseni Mountains (Pătraşcu et al., 1990) and Banat (this study) are *in situ*, and those from the Haţeg and Rusca Montană basins (Pătraşcu and Panaiotu, 1990) are after tectonic correction.

According to Berbeleac (1988) these changes are produced by the associated metasomatic and hydrothermal systems during the cooling of the bodies between 800°C and 200°C. Taking this and the absence of metamorphic or volcanic events in the Banat area during the Tertiary (Mutihac and Ionesi, 1974) into account, we think that the age of the secondary magnetic minerals is also Late Cretaceous and, consequently, the remanent magnetizations carried by them have the same age.

Field and laboratory methods

In the field we have collected 148 block-samples from 44 sites. The block-samples were oriented using a sun compass and, in the absence of sun, using a magnetic compass and a declination gradiometer (Doell and Cox, 1962). We cannot determine whether the intrusions underwent tilts about horizontal axes, because in our sampling area there were no stratiform beds slightly younger than the intrusions deposited directly on them. Therefore, we have assumed that these post-tectonic intrusions were not significantly tilted, as is usually accepted in such cases.

In the laboratory, a minimum of four cubic specimens (13.8 cm^3) from each sample were prepared. The natural remanent magnetization (NRM) of specimens was measured using an astatic magnetometer LAM 24 using As' method with 24 positions (As, 1960, 1967). The structure of the NRM was analysed by alternating field (AF) demagnetization, using a two-axis tumbler placed within the field-free space provided by a Helmholtz coils system capable of maintaining a reduced field of 10 nT or less (Pătraşcu, 1976). One to four specimens from each sample were demagnetized in 10–12 steps up to 100 mT, and demagnetization data were plotted using Zijderveld's method (1967). Some demagnetization steps were repeated to check for consistency. The

other specimens from each sample were demagnetized in one to four steps to isolate the characteristic magnetization of the sample. In all the cases, the field used for that purpose was greater than 20 mT.

Samples were examined petrographically and mineralogically in reflected and transmitted light.

Results

Rejected sites

At most of the sites, the intensity of NRM of specimens decreased by 80–90% on demagnetization at 20 mT, and when we begin to continue the demagnetization up to 100 mT the directions of magnetization become erratic, preventing us from identifying any components between 20 and 100 mT. The magnetic directions of components isolated up to 20 mT are different from sample to sample, and sometimes even from specimen to specimen. The directions of some of the samples are similar to the present geomagnetic field at the sampling sites. All of these soft components are probably field or laboratory viscous remanent magnetizations. Specimens from several sites failed to yield consistent magnetic directions after repeated partial demagnetization at the same peak field greater than 20 mT, indicating that no observable component of magnetization was present. All these specimens have Q ratios less than 0.5, the average being around 0.2. These specimens are from samples affected by metasomatic or hydrothermal processes.

In the Bozovici area one site yielded internally consistent paleomagnetic data. The hard component of magnetization was isolated on demagnetization at 100 mT and has a direction similar to the present geomagnetic field at the sampling site. Taking the petrographic evidence of alteration of these samples into account, this component is probably a secondary magnetization.

34 of the total of 44 sites were rejected for the reasons just outlined.

Stable and consistent sites

The remaining ten sites (Table 1, Banat area) were found to be magnetically stable and to display intra-site consistency in their magnetic directions. Most of the specimens have an NRM composition comprising two components, a soft component destroyed in peak fields up to 0.2–20 mT, and a hard component. The soft components, either with chaotic directions or with directions similar to the present geomagnetic field direction, are probably viscous remanent magnetizations. Site-mean directions and paleomagnetic poles for these ten sites are presented in Table 1. In the same table the other paleomagnetic data from the Romanian sector of the Apuseni–Srednogorie magmatic belt, after Pătraşcu et al. (1990) and Pătraşcu and Panaiotu (1990) are also present. The site-mean data were computed using Fisher's statistics (1953) as follows: the hard magnetization directions of specimens were combined to obtain a mean direction for each oriented sample and the latter were combined to obtain site-mean directions. For the sites where the small outcrops did not allow us to collect more than two samples, the specimen directions were combined to obtain the site-mean.

These ten sites are from the following areas.

1. Dognecea

The two sites (D-5 and D-13) were collected from unaltered granodiorites and granites. Small magnetite grains are present as idiomorphic inclusions in plagioclase, and the samples show a rock structure typical of fast cooling in agreement with the marginal position of these sites in the Dognecea body. The hard components from the two sites had a normal polarity and their initial intensities were reduced by 75–80% on demagnetization in 75 mT (Fig. 3a). They are probably thermoremanent magnetizations acquired by magnetite grains during the cooling of the body.

2. Bocşa

There are three megadykes (B1, B2 and B3) in the Bocşa Massif, which cut the metamorphic formations of the Supragetic Units. The eastern type (B3) is believed to have been intruded after B1 and B2 (Russo-Săndulescu et al., 1978). Consistent paleomagnetic data were obtained from B2 (sites B2-11, B2-18) and B3 (sites B3-27, B3-2, B3-30, B3-31). Samples from sites B2-11 and B2-

18 are quartz monzonites and granites. Petrographic examination of site B2-18 reveals that the quartz exhibits undulatory extinction on a mosaic structure, plagioclase has mechanical macles and most of the crystals are broken. It is quite probable that these rocks have undergone a dynamic stress, produced by the emplacement of a younger intrusion (B3). The small magnetite grains are present as inclusions in cloudy feldspars. The hard component of the remanent magnetization is of normal polarity at both sites, and it was reduced by 50–60% in a peak field of 100 mT (Fig. 3b). These rocks have not suffered extensive alteration produced either by the later intrusion (B3) or by associated pervasive hydrothermal systems, but a dynamic perturbation by the intrusion of B3 is possible. The amplitude of this perturbation could not be estimated by geological arguments alone, and taking into account the paleomagnetic data (Fig. 4 and Table 1), this perturbation is probably associated with a minor tectonic movement.

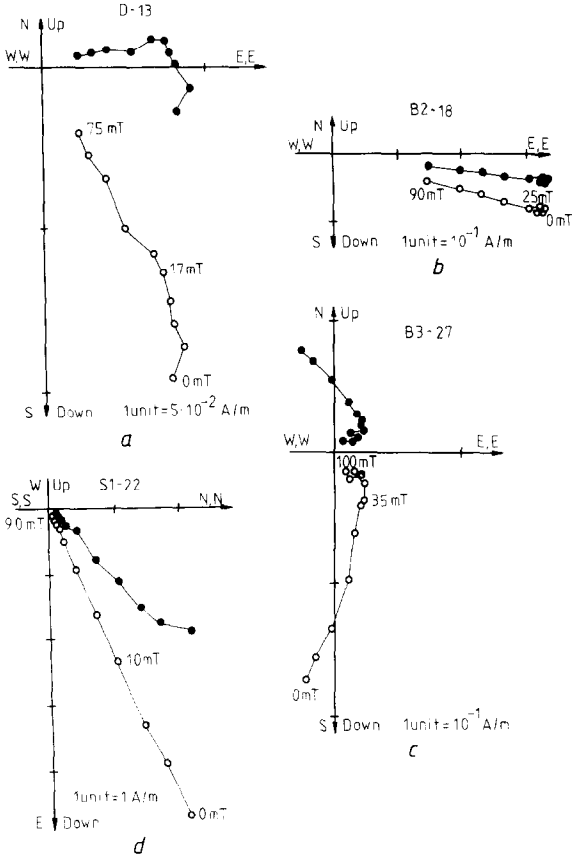


Fig. 3. AF demagnetization plots of samples from Banat sites: site numbering according to Table 1. Open circle = vertical projection; full circles = horizontal projection.

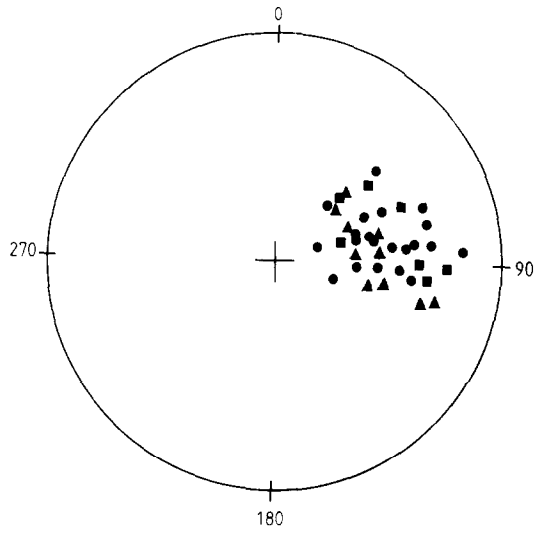


Fig. 4. Stereographic projections of site-mean directions from Banat (triangles), the Apuseni Mountains (circles) and the Haţeg and Rusca Montană basins (squares). The site-mean directions of sites with reversed polarity (see Table 1) are inverted through the origin.

The sites from B3 with consistent results consisted of monzonites (B3-27 and B3-30) and quartz granodiorites (B3-2 and B3-31). All of the samples show petrographic evidence of alteration. The plagioclase is cloudy (altered to calcite and muscovite at B3-27 and B3-30), green amphibole and biotite are replaced by chlorite, epidote, sphene and hematite, and magnetite is oxidized to hematite. The specimens have one or two components of magnetization which are destroyed in peak fields up to 25–30 mT, with chaotic directions or aligned to the present geomagnetic field, and a hard one which displayed greater resistance to AF demagnetization than the hard components from the sites which have not been affected by metasomatic processes (Fig. 3c). The metasomatic alterations observed in our samples were acquired during the cooling between 800°C and 200°C (Berbeleac, 1988). As the creation and alteration of magnetic phases most probably continued during cooling below their respective Curie temperatures, the remanence carried by the altered samples may be a ther-

moremanence together with a thermochemical remanence, both of which probably carry the hard remanence, and their contributions cannot be separated by AF demagnetization. The susceptibility at these four sites is less than the average value of susceptibility for the other sites of B3 with metasomatic alterations, which were rejected owing their weak resistance to AF demagnetization.

3. Surduc

This area is made up of two successive intrusions: the first (S1) with weak alkaline character and the second (S2) with calc-alkaline character (Russo-Săndulescu et al. 1986).

Gabbro-diorites were collected from site S1-22. Mineralogical studies show that the pyroxene contains exsolved opaque iron oxides, which are often more abundant in the marginal parts of the crystals, and some of which are partially replaced by secondary green amphibole; olivine has been pseudomorphed by talc, magnetite and chlorite; titanomagnetite is converted to ilmenite, some rutile and Ti-poor titanomagnetite; and hydrothermal alterations are absent. The hard remanent magnetization has normal polarity and the AF demagnetization in 100 mT reduced its intensity by 90% (Fig. 3d). A similar petrographic structure was observed by Dagley et al. (1990) in Scotland and was interpreted as a metamorphic effect, due to a later intrusion. It is likely, therefore, that this site has been completely magnetically overprinted by the later intrusion S2.

Site S2-21 was sampled in the body S2 from an altered granodiorite, with hematite as the dominant magnetic phase. The NRM composition is similar to that observed in B3. The direction of the hard component obtained is close to that of S1-22 in the nearby S1 body.

Taking into account that the magnetic minerals identified in our samples were formed during the cooling of the bodies, the age of the characteristic magnetization is the same as that of the bodies, i.e. Campanian–Maastrichtian. Unfortunately, the regional geological structure has not allowed us to perform field tests to constrain the age of the characteristic magnetization better. However, looking to the assemblage of Campanian–Maastrichtian paleomagnetic data from the Apuseni Mountains, the Haţeg and Rusca Montană basins and Banat (Table 1, Figs. 2, 4, 5), we conclude that: (1) the tilt corrected site-mean directions from the Haţeg and Rusca Montană basins are in agreement with the *in situ* site-mean directions from the Apuseni Mountains and Banat; (2) the reversed magnetizations are antiparallel to the normal polarity (positive reversal test with classification C according to McFadden and McElhinny's (1990) reversal test); (3) the 95% confidence circles of the paleomagnetic poles overlap for the three areas; and (4) the site-mean directions are different from the direction of the present geomagnetic field. All of these considerations provide sufficient arguments for us to accept the primary character of these magnetizations.

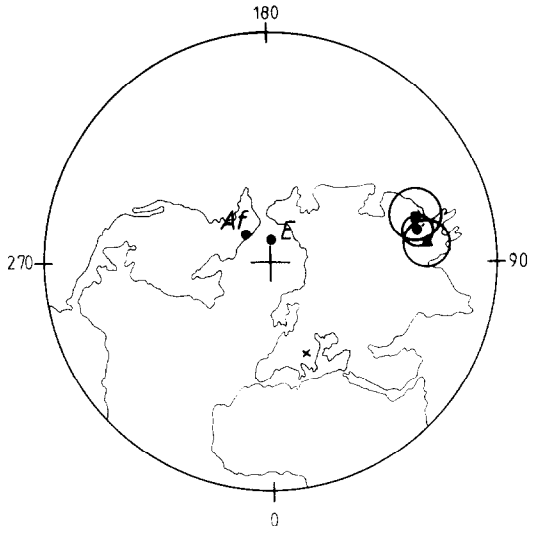


Fig. 5. Stereographic plot of paleomagnetic poles and their 95% confidence circles from the Apuseni Mountains (circle), the Haţeg and Rusca Montană basins (square) and Banat (triangle). Coeval paleomagnetic poles for Europe (*E*) and Africa (*Af*) are after Westphal et al. (1986, Table 4). The location of the Apuseni–Banat area is marked with cross.

We also remark that the sampled sites in the Apuseni Mountains have only reversed polarity. South of Mureş river, in the Southern Carpathians, both normal and reversed polarity, with a strong bias toward normal polarity, were found. Our samples have been collected from different volcanic and intrusive rocks and are spread over relatively large areas. Taking into account these

remarks, we think that the cooling and magnetization of the studied rocks took place during a period of time characterized by long intervals (more than 1 Ma) with the same geomagnetic polarity. This is in agreement with the age of these magmatic rocks, Campanian–Maastrichtian, because the Campanian in particular is characterized by intervals that satisfy this condition (Lowrie and Alvarez, 1981).

Implications for tectonics

Because the 95% confidence circles of the paleomagnetic poles from the Apuseni Mountains, the Haţeg and Rusca Montană basins and Banat overlap, we cannot deduce any significant latitudinal movement or rotation between these areas after the Cretaceous, so that the paleomagnetic data indicate that the Romanian sector of the Apuseni–Srednogorie magmatic belt moved like a rigid block in post-Cretaceous time. The relationship with stable Europe, stable Africa and some adjacent zones of the Tethys mobile belt will be analysed by considering a mean paleomagnetic pole for the three regions. Since these regions are separated by only several tens of kilometers, the overall mean computed from the three area-means will provide a more realistic estimate of the paleomagnetic pole than by treating the 38 sites as independent data (Table 1).

To interpret the data in terms of tectonic evolution, this paleomagnetic pole must be compared with the coeval paleomagnetic poles of stable Europe and stable Africa, which have been extracted from Table 4 of Westphal et al. (1986). The rotation (R) and the flattening (F) parameters, calculated by equations given by Beck (1980) and modified by Demarest (1983), are used to obtain the tectonic significance of the data (Table 2).

TABLE 2

Concordance/discordance statistics for Apuseni–Banat

	Africa	Europe
R	$86.0° \pm 10.6°$	$73.0° \pm 11.2°$
F	$7.7° \pm 12.3°$	$15.9° \pm 11.2°$

These parameters show a large clockwise rotation both with respect to Europe and Africa and a northward transport with respect to Europe, but no significant latitudinal movement with respect to Africa.

Our paleomagnetic data impose some constraints for the tectonic models of this part of the central Mediterranean during Upper Cretaceous and Tertiary time.

(1) The Upper Cretaceous magmatic rocks from Apuseni–Banat were emplaced in a block situated to the south of Europe. The observed fossil inclination has an African affinity, but the evolution of the positions of this block with respect to Africa cannot be established in detail, owing to the absence of a polar wander curve for this block through the Mesozoic and Tertiary.

(2) The fossil declination observed on the Upper Cretaceous magmatic rocks from Apuseni–Banat is similar to that observed on some remagnetized Mesozoic rocks from the northern Apuseni Mountains (Surmont et al., 1990) and with those observed in Cretaceous and Tertiary rocks from the Mecsek Mountains (Márton, 1986). Taking this into account we can define a paleomagnetic unit characterized by a large clockwise rotation which took place during the Cenozoic in the southern and eastern parts of the Carpatho-Pannonian realm, in contrast with the northwestern region, where counter-clockwise rotations are dominant (Fig. 1). The limits of this unit are difficult to trace using the present paleomagnetic data. The NE–SW limit, according to the paleomagnetic results from the Pannonian Basin and the Carpathians, is probably along the mid-Hungarian Belt (Márton, 1987; Márton and Mauritsch, 1990). Part of the southern limit is probably located somewhere in eastern Serbia, because the rotations are barely significant or absent to the south and east of the Timok river (Nozharov and Veljovic, 1974, 1984; Nozharov and Petkov, 1976; Nozharov et al., 1977, 1987; Veljovic et al., 1980; Veljovic and Nozharov, 1981; Kruczyk and Kadzialko-Hofmokl, pers. commun., 1990).

(3) In Figure 6 we present the inclinations observed on Senonian rocks from the northwestern part of Carpatho-Pannonian domain, characterized by counter-clockwise rotations (CCW);

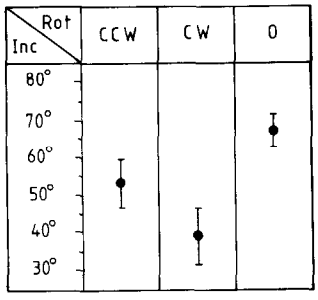


Fig. 6. Senonian average paleomagnetic inclinations and their 95% confidence intervals for areas with different rotations. Reference paleomagnetic data: counter-clockwise rotated area (*CCW*) = Bakony area (Márton and Márton, 1983); clockwise rotated area (*CW*) = Apuseni–Banat; no rotation area (*O*) = the Southern sector of the Apuseni–Srednogorie magmatic belt (see text).

those from the eastern part of the Carpatho-Pannonian domain, characterized by clockwise rotations (*CW*); and those from the southern sector of the Apuseni–Srednogorie magmatic belt, with no significant rotation (*0*). For the northwestern unit we have plotted the paleomagnetic data from the Maastrichtian sediments of the Bakony area (Márton and Márton, 1983) and for the southeastern unit the data from the Campanian–Maastrichtian magmatic rocks of the Apuseni–Banat area. For the southern sector of the Apuseni–Srednogorie magmatic belt we used the paleomagnetic data from Nozharov and Petkov (1976), Nozharov et al. (1977, 1987), Nozharov and Veljovic (1984) and Veljovic and Nozharov (1981). After the elimination of intermediate or abnormal directions, we have selected only the results from the sites with $k > 10$ and $\alpha_{95} < 16$ for the tectonic interpretation. Combining the 20 selected sites (18 sites with normal polarity and two sites with reversed polarity) we have obtained this area-mean direction for the southern sector of the Apuseni–Srednogorie magmatic belt: $D = 0.8°$, $I = 65.2°$, $\alpha_{95} = 5.5°$, $k = 35$. Application of the central axial dipole hypothesis allow us to draw the following conclusions regarding the relative positions in latitude between these units with different rotations during the Senonian for the data presented in Fig. 6: (1) the northwestern part of the Carpatho-Pannonian domain was situated north of the Apuseni–Banat area; (2) the southern sector of the Apuseni–Srednogorie magmatic belt (between the Timok river and the Black Sea) seems to have been situated a great distance to the north of its western part (Apuseni–Banat). The paleomagnetic data from the southern sector are consistent with a position on the southern margin of Europe. These paleomagnetic data are not compatible with a continuous magmatic belt from Apuseni to the Black Sea during the Senonian, nor with a common origin for these magmatic rocks. However, we must remark that the area-mean direction of the southern sector of the magmatic belt is similar to that of the present geomagnetic field in that area, and the age of magnetization is not very well constrained by field or laboratory tests. Therefore, these paleomagnetic data need to be scrutinized further before a definitive interpretation can be made.

Conclusions

The paleomagnetic results from the western part of the Southern Carpathians (Banat area) reported here yield new evidence for a rigid connection between the areas penetrated by the Upper Cretaceous magmatic rocks in Romania during the Cenozoic. These magmatic rocks, together with the tectonic units penetrated by them, were situated in a block south of Europe during the Senonian. At the same time, the northwestern part of the Pannonian Basin was situated north of the Apuseni–Banat area. These paleomagnetic data from the Apuseni–Banat area, together with other data from the Carpatho-Pannonian domain, have allowed us to define a unit characterized by a large clockwise rotation during the Tertiary in the eastern and southern parts of this domain.

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